MILITARY SPECIFICATION
ELECTRONIC SHOP, SHELTER MOUNTED
AN/MSM-108( )

This specification is approved for use within Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. - This document covers the following:

1.1.1 Description. - Electronic Shop, Shelter Mounted AN/MSM-108( ), hereinafter referred to as an Electronic Shop, is a mobile assemblage housed in a modified S-280( )/G shelter equipped with work benches, storage cabinets and racks, lighting, air conditioning, and power to form an electronic repair shop for the maintenance of electronic equipment. It is capable of being transported by railroad, military truck and airlift.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, New Jersey 07703-5016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-3885	Cable Assemblies and Cord Assemblies (for use in Electronic and associated Equipment)
MIL-W-6858	Welding, Resistance Aluminum, Magnesium, Non-Hardening Steels or Alloys, Nickle Alloys, Heat Resisting Alloys, and Titanium Alloys, Spot and Seam
MIL-I-23053/6	Insulation sleeving, electrical, heat shrinkable,polyolefin, semirigid cross-linnked
MIL-T-7928	Terminals, Lug and Splice, Crimp Style, Copper
MIL-W-8611	Welding, Metal Arc and Gas, Steels and Corrosion and Heat Resistant Alloys, Process for
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-W-45205	Welding, Gas Metal Arc and Gas, Tungsten-Arc, Aluminum Alloys
MIL-S-55507	Shelter, Electrical Equipment (with or without equipment) packaging of

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STANDARDS

MILITARY

	MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
	MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
	MIL-STD-490	Specification Practices
	MIL-STD-454J	Standard General Requirements for Electronic Equipment
	MIL-STD-1472C	Human Engineering Design Criteria
	MIL-STD-1474B(MI)	Noise Limits for Army Materil
	MIL-STD-810D	Environmental Test Methods
DRAWINGS	DL-A3022630	Electronics Shop Shelter mounted, AN/MSM-108
	A3023000	Information Plate, Power Schematic

2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.

2.2 Order of precedence. In the event of a conflict between the test of this specification and the references cited herein (except for associated detail specification, specification sheets or MS standards), the test of this specification shall take precedence. Noting in this specification, however, shall supersed applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. - The contractor shall furnish an unit of the AN/MSM-108( ) in accordance with 4.3.

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3.2 Construction. - (See 3.23)

3.2.1 Shelters. - Unless otherwise specified the Shelter, Electrical Equipment S-280( )/G will be Government furnished.

3.2.2 Dimensions and Weight - The S-280 ( )C/G shelter has the following dimensions and weight:

	Height	Width	Length
Exterior	86 3/8"	87"	147"
Interior	77 1/8"	81 1/2"	138"

Weight: 1390±80 lbs. excluding the MK-1079( )C/G

3.2.3 Electronic shops installation. -

Electronic Shop, Shelter Mounted AN/MSM-108( ) shall be constructed in accordance with the drawings listed on DL-A3022630 of this document and the Statement of Work for the electronic shop, shelter mounted AN/MSM-108 issued on 6 Sep 1984.

3.3 Electrical installation. (See 4.6)

3.3.1 Power and signal system wiring and components. - The completed in-place power distribution system and signal wiring shall meet the following requirements:

3.3.1.1. Continuity of conductors. - The point-to-point resistance between each pin of the power and signal entrance connectors and respective circuit terminations shall be less than 1 ohm. (See 4.6.1.1, 4.6.2)

3.3.1.2 Dielectric breakdown and insulation resistance. - All power and signal wires, cables and components shall withstand without flashover or insulation breakdown, a potential of 500 volts DC between each of the ungrounded power line conductors and shelter facility ground applied continuously for not less than thirty (30) seconds and the insulation resistance shall not be less than 100 meagaohms when tested in accordance with 4.6.1.2, 4.6.1.3, 4.6.2.

3.3.2 Power distribution and operational functions. - Main power, branch distribution and circuit switching functions thereof shall meet the operational requirements of power schematic diagrams A3023000 when tested in accordance with 4.6.1.4

3.3.3. Cable assemblies, power and audio frequency. - All power and audio frequency cable and cord assemblies shall be fabricated in accordance with

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the specified requirement and meet the continuity, dielectric, insulation resistance and pull-out requirements of MIL-C-3885. (See 4.21)

3.4 Vehicular transportation- The shelter assemblage shall be capable of being transported over cross-coutry terrain by military vehicle without sustaining any damage such as buckling, cracking, delamination, or other permanent deformation (See 4.8). A 5 ton truck, for example the M-923 will be used. MIL-STD-810C Method 514.2, Procedure 13, Part II shall be implemented.

3.5 Railroad transportation. - The shelter assemblage shall be capable of being loaded, blocked and braced on a flatcar for shipment by rail and shall be subjected to and withstand the tests of 4.9 without sustaining any permanent damage such as buckling, cracking, delamination or other permanent deformation.

3.6 Drop test. - The shelter assemblage shall be capable of withstanding drops of eighteen (18) inches onto concrete without sustaining any permanent damage such as buckling, cracking, delamination or other permanent deformation. (See 4.10).

3.7 Humidity Test. - The equipment (including shelters) shall be operable without degradation in specified performance, after tested in accordance with humidity test of MIL-STD-810D, method 507.2 procedure II for ground and airborne electronic equipment.

3.8 Air tightness test. - The shelter facility shall show no evidence of air leakage through seams, joints, rivets, etc, or modified exterior surfaces when tested for tightness in accordance with 4.11.

3.9 High and low temperature - The equipment shall be operable without degradation in specified performance after tested in accordance with 4.7. The temperatures at which the equipment shall perform without degradation will be: Ambient(-46°C)-50°F to (49°C) 120°F and storage/transportation. (-54°C)-65°F to (68°C) 155°F.

3.10 Mounting inserts. - All mounting inserts, unless otherwise specified, shall be installed without the use of any auxiliary materials such as adhesive and, when installed, each insert shall be capable of withstanding the tensile load and the torque load referenced in 4.13. There shall be no damage or permanent distortion to the shelter mounting surface or insert head and the threads and the inserts shall not have rotated from their original position. Insert threads shall be free of paint.

3.11 Acoustical noise. - The acoustical noise shall not exceed 75dB(A) in accordance with the criteria specified in MIL-STD-1474(MI), Table 2, Category E (See 4.20).

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3.12 Cleaning - After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in the applicable portions of MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, scale, and any other foreign material. In addition, when necessary, such cleaning shall also be performed before and after assembly of the units. (See 3.24).

3.13 Finish. - The shelter facility shall be finished in accordance with MIL-F-14072 and the equipment drawings (See 3.24 and 4.4).

3.14 Marking. - (See 3.24 and 4.4.)

3.14.1 General - Marking shall conform to MIL-M-13231 and the equipment drawings.

3.14.2 Nameplates. - Shelter facility and assemblage shall be provided with a nameplate in accordance with the equipment drawings. Each nameplate shall be imprinted with a sequential serial number.

3.14.3 Aircraft loading data plate. - The shelter assemblage shall be provided with data plates in accordance with equipment drawings, imprinted with the dimensions which locate the center-of-gravity of the shelter facility and the assemblage in the three planes; the gross weight of the shelter facility and assemblage; and the bearing pressure on the bottom of the skids. The location of the centers-of-gravity shall be determined by the contractor using the test article/unit of production.

3.14.4 Interior and exterior marking. - All interior and exterior markings shall be as defined by the equipment drawings unless otherwise specified.

3.15 Riveting - Riveted joints shall be tight; the joined parts shall be undamaged; and the rivet heads shall be properly seated and tight against their bearing surfaces. Rivets shall conform to requirement 12 of Para. 6 of MIL-STD-454. (See 3.24 and 4.4).

3.16 System Safety Engineering

3.16.1 Personnel Hazards. -Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1 and 8 shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly process, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual inspection

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(See 4.24).

3.16.2 Edge Rounding Exposed. - Exposed edges shall be rounded to a minimum radius of 0.04 inch (1mm) and exposed edges corners to a minimum of 0.05 inches (1.3 mm).

3.16.3. Radioactive Material. - Radioactive materials shall not be used (E.G. luminous dials/marking, electrontubes, surge arrestors and lenses).

3.17 Seal, water vapor. - All joints and edges of the framing for the exhaust blower, power and antenna entrance boxes and air-conditioner shall be sealed to provide a barrier against the entrance of water vapor to the core material of the shelter panels and to the interior of the shelter. All joints and edges assembled and sealed in a manner to prevent collection and retention of moisture. All exterior mounting inserts, rivets, screws, and fasteners shall be coated by a one-part sealant Silastic 738, as made by Dow Corning, or equal, using the manufacture recommended primer. (See 3.23).

3.18 Securing of parts. - Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall retain components of the facility securely so that loosening, wear of mounting arrangements or permanent separation of parts or components will not occur when the equipment is subjected to the specified drop and transport tests. (See 3.23 and 4.17)

3.19 Soldering. - (See 3.23 and 4.17)

3.19.1 Solder and flux. - Solder and flux shall be as defined by equipment drawings.

3.19.2 Acid or acid salts. - No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.19.3 Process . - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign materials remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation

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material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.20 Tropicalization of material. - Canvas, webbing, wood, etc. shall be treated to be resistant to moisture an fungi as follows:

3.20.1 Treating materials. - Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.20.2 Toxicity. - Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.20.3 Flexibility. Treatment shall not affect the flexibility of treated materials to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.20.4 Statement of treatment. - The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use in case the drawings do not specify the teatment in detail (See 4.22)

3.21 Welding. - Welding joints shall be as defined by the drawings and shall be such that grinding on the finished weld will be unnecessary except where grinding is specified on the drawings. Inert-gas-shielded arc welding using helium, argon, or a mixture of the two shall be used for arc welding of aluminum. Welds shall conform to MIL-W-6858 CL.B, MIL-W-45205 CL.B or MIL-W-8611 as applicable . All surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free of scale, paint, grease, and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free of scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely neutralized and removed. (See 4.4)

3.22 Wiring and cabling. - (See 3.23 and 4.4) Wiring and cabling shall be in accordance with the drawings and shall be neat and sturdy.

3.22.1 Slack. - Wires and cables shall be as short as practical except that sufficient slack shall be provided:

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(a) To prevent undue stress on cable forms, wires, and connectors.

(b) To permit removal from rack of operational equipments, having wiring or cables to connectors at back of equipment, so that cable may be connected or disconnected.

(c) To facilitate field repair of broken or cut wires.

(d) To prevent chafing or breaking of wires due to repeated flexing of hinged parts.

3.22.2 Protection. - Wires and cable shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation materials.

3.22.3 Splicing. - Wires in a continuous run between two terminals shall not be spliced during the wiring operation unless specifically indicated by the equipment drawings.

3.22.4 Solderless terminal lug connections. - Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements and tests of MIL-T-7928. (See 4.4 and 4.15)

3.22.5 Identification of wiring. - Wiring shall be identified by color coding and wire pair marking as indicated on the drawings.

3.22.6 Testing. - Power and signal wiring shall be subjected to and pass the tests described in 4.6.

3.23 Interchangeability. - Like units, assemblies, sub-assemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items of the equipment (See 4.16). Individual items shall not be handpicked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc. Provisions for interchangeability shall be in accordance with requirement 7 of MIL-STD-454J.

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3.24 Workmanship. - (See 4.17) The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454J and the applicable portions of the following paragraphs:

3.2	Construction
3.3	Electrical installation
3.10	Mounting inserts
3.12	Cleaning
3.13	Finish
3.14	Marking
3.15	Riveting
3.17	Seal, water vapor
3.18	Securing of parts
3.19	Soldering
3.20	Tropicalization
3.21	Welding
3.22	Wiring and cabling

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - The contractor is responsible for the performance of all inspection requirements as specified herein. The contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by CECOM. CECOM reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 CECOM verification. - All quality assurance operations required of The contractor shall be subject to CECOM verification at scheduled or unscheduled intervals. Verification will consist of the following:

Surveillance of the contractor's operations to determine that practices, methods and procedures of the contractor's product assurance system are being applied, and are in fact accomplishing the objectives of total compliance of the product with the requirements of this document.

4.1.2 Accommodation and Assistance. - The contractor shall make his inspection equipment available for use by CECOM quality assurance representative for verification purposes. The contractor's personnel shall be made available for operation of such inspection equipment as required. All facilities and services necessary for the placement operation and maintenance of these test requirements shall be provided by the contractor.

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4.2 Classification of inspections. - Inspections shall be classified as follows:

(a) First Article Inspection (See 4.3) does not include packaging.

(b) Inspections covered by subsidiary documents. (See 4.4)

(c) Quality conformance inspection of equipment before packaging. (See 4.5)

(d) Packaging inspection. (See 4.18.)

4.3 First Article. - The First Article inspection shall be performed by contractor. (See 3.1)

4.3.1 First Article Inspection. - This inspection shall consist of the inspections specified in subsidiary documents covering the items listed in paragraph 4.4 and the inspections specified for group A, group B and group C (see Tables II, III, IV, respectively). The inspections shall be performed in the order shown in Table I.

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TABLE I

First Article Inspection

Inspection	Requirement Paragraph	Test Para
Tropicalization of material*	3.20	4.22
Interchangeability	3.23	4.16
Solderless-terminal-lug connections	3.22.4	4.15
Mounting Inserts	3.10	4.13
Subsidiary documents	3.13,3.14,3.17.3.21,3.22	4.4
Visual and mechanical	3.24	4.17
Electrical installation	3.3	4.6.1/4.6.2
Vehicular transportation	3.4	4.8
Railroad transportation	3.5	4.9
Drops, flat and rotational	3.6	4.10
Air tightness	3.8	4.11
Acoustical noise	3.11	4.20
Visual and mechanical	3.24	4.17
Electrical installation	3.3. 1, 3.3.2	4.6.1/4.6.2
High and low temperature	3.9	4.7
Humidity	3.7	4.14

Inspection marked (*)required only when materials or processes other than those defined by this document are used.

4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery, required by this document, and the inspection requirement specified in the contract or purchase order:

ITEM	WHERE REQUIRED
Finish	3.13
Marking	3.14
Riveting	3.15
Welds	3.21
Wiring and cabling	3.22

4.5 Quality conformance inspection of equipment before packaging. - The contractor shall perform the inspection specified in 4.4 and 4.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to

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assure compliance with all document requirements. CECOM will review and evaluate contractor's inspection procedures and examine contractor's inspection records as an element of CECOM verification. (See 4.1.1)

4.5.1 Group A inspection. - Each unit on contract shall be inspected for conformance to the inspections specified in Table II. Lots shall be formed only from units that pass this inspection. Factors of lot composition not defined herein, or in the contract shall be in accordance with MIL-STD-105. Each lot may be subjected to a verification audit, utilizing the procedures of MIL-STD-105, using the general inspection levels and the AQLs indicated in Table II".

TABLE II

GROUP A INSPECTIONS

Inspection	Reqt Para	Insp Para	AQL% Major	AQL% Minor
Visual & Mechanical MSM-108	3.24	4.17	1.5	6.5
Airtightness	3.8	4.11		
Electrical Installation MSM-108	3.3	4.6.	No failures Allowed	

4.5.1.1 Order of inspection within Group A. - Group A inspection shall be performed in the order listed in Table II.

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4.5.2 Group B inspection - This inspection shall conform to Table III and to the special inspection level S-4 of Table I of MIL-STD-105. Group B inspection shall be performed on units that have passed Group A inspection.

TABLE III

Group B Inspection

Inspection	Reqt Para	Insp Para	AQL%
Mounting inserts	3.10	4.13	4.0
Interchangeability	3.23	4.16	4.0

4.5.2.1 Order of inspection within Group B. - Group B Inspection shall be performed in the order listed in Table III.

4.5.3 Group C Inspection. - Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table IV.
Samples shall be selected in accordance with 4.5.3.1 thru 4.5.3.2. Group A inspection shall be repeated after completion of Group C inspection.

TABLE IV

Group C Inspection

Inspection	Reqt Para	Insp Para
Mounting inserts	3.10	4.13
Solderless-terminal lug connections	3.22.4	4.15
Vehicular transportation	3.4	4.8
Railroad transportation	3.5	4.9
Drops, flat and rotational	3.6	4.10

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4.5.3.1 Sampling for inspection of solderless-terminal-lug-connections. - Five specimens of each combination of wire and terminal produced each month shall be prepared for inspection. A specimen is a one foot length of wire on which a terminal lug has been applied using the same tools and procedures used in crimping the terminal lugs in production.

4.5.3.2 Sampling for other Group C Inspection. - For inspection tests listed below, one unit shall be randomly selected from the first 10 units (or fraction thereof) produced. Thereafter, one from each additional 150 units (or fraction thereof) produced shall be subjected to the below listed tests Selection of the unit will be made by CECOM.

a. Mounting inserts
b. Acoustical noise
c. Humidity
d. Tropicalization of materiel
e. Railroad transportation
f. Drops, flat and rotational
g. Air tightness
h. Vehicular transportation
i. High and low temperature

4.5.3.3 Order of inspection. - Group C inspection shall be performed in the order listed in Table IV.

4.5.3.4 Non-compliance. - All quality conformance inspection shall be halted, including group A and group B inspections, upon occurrence of any group C failure. The contractor shall immediately report, in writing, each group C failure occurence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on the processes and materiels, as applicable and on all units of production which are manufactured under the same conditions and which CECOM considers subject to the same failure. Reports shall be forwarded to CECOM Procurement Quality representative and to the responsible technical activity designated in the contract.

4.5.3.5 Reinspection of conforming group C sample units. - After corrective action has been taken, the failed unit and /or additional sample units shall be subjected to group C inspection (all inspections, or the inspections which the sample failed, at the option of the technical activity). Any or all group A inspections may also be reinstituted. All damage resulting from any testing shall be repaired. Final acceptance and shipment will be withheld until the reinspection results have shown that the corrective action was effective and approved by the technical activity.

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4.6 Electrical installation tests. (See 3.3) The tests of 4.6.1 or 4.6.2 as applicable shall be performed on each shelter facility. Before applying power to the shelter facility an electrical ground at the power entrance box shall be made. Power to be supplied shall be 120/208 volts A.C., 3 phase, 60 Hz.

4.6.1 Power system wiring and components installation tests - Prior to conducting the tests of 4.6.1.1 thru 4.6.1.3 below, the following shall be accomplished (Refer to the power wiring diagram).

(a) Remove or disconnect the following loads:

(1) All fluorescent, incandescent and neon (Power indicator, power distribution indicator, and correct-incorrect phase indicator) lamps.

(2) All fluorescent lamp starters.

(3) Heater, D. C. power supplies, and air conditioners.

(b) Remove power distribution box cover panel and disconnect the 2 conductors from the voltmeter, frequency meter terminals and ammeter. Position or isolate disconnected leads so as not to short to each other or to shelter facility ground.

4.6.1.1 Continuity of conductors. - Continuity resistance tests shall be performed on each power circuit in accordance with the following paragraphs. Continuity shall exist between each pin of the power entrance connector and respective circuit termination and shall be less than 1 ohm. A calibrated low resistance reading ohm meter or equivalent shall be used in performing the test.

(a) Place the following controls in the ON position.

(1) Fluorescent light switches

(2) Incandescent cold start switch

(3) All circuit breakers

(b) Place the Blackout Bypass switch in the Bypass position.

(c) Measure continuity resistance as follows:

FROM	TO
Pin A of Power Input Connector at the power entrance box	Black wire Termination at fluorescent and incandescent lamp

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	Black wire Termination at air conditioners 1 and 2
	Black wire Termination of all 60Hz convenience receptacles on roadside walls.
	Black wire termination (brass contact) of heater receptacle and air compressor.
Pin B of Power Input Connector	Red wire Termination at air conditioners 1 and 2
	Red wire termination of D. C. power supply I.
	Red wire termination (brass contact) of all 60 Hz convenience receptacles on curbside walls.
Pin C of Power Input Connector	Blue wire termination at air conditioners 1 and 2
	Blue wire termination of DC power supply 2.
Pin N of Power Input Connector	White wire terminations at each fluorescent and incandescent lamp, air conditioners 1 and 2, D. C. power supplies, all convenience receptacles, heater receptacle and air compressor receptacle .
Pin G of Power Input connector	Ground Terminal E1, and ground contact of all electrical outlets 60Hz
Positive terminals of D. C. supplies	Each positive binding post: to universal binding post for power supply I.

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	To AN/VRC-46 radio binding post for power supply 2.
Negative terminals of D. C. supplies	Each negative binding post: to universal binding post for power supply 1. To AN/VRC-46 radio bindig post for power supply 2.

d. No continuity shall exist between the following:

(1) Each pin of power entrance connector and each of the remaining pins

(2) Positive and negative D. C. binding posts

4.6.1.2 Dielectric Strength of insulation. - Dielectric Strength Tests shall be performed on each power line circuit in accordance with the following paragraphs. The insulation of all power wires, cables and components shall withstand, without flashover or insulation breakdown, a potential of 500 volts D.C. applied continuously for not less than 30 seconds.

NOTE: Take preventive measures that will preclude personnel from coming into contact with circuits under test during performance of all dielectric strength and insulation resistance tests. Discharge each conductor to ground following each test in order to eliminate danger of electric shock.

(a) Retain all circuit controls tested under 4.6.1 above in the ON position except fluorescent lamp switches.

(b) Disconnect at the meter switch all lines to the phase indicator circuitry. Isolate the disconnected lines so as not to short to each other or to shelter facility ground.

(c) Test between pin A of the power input connector and all other pins of the power input connector strapped to shelter facility ground. Adjust tester for an output of 500 volts D.C. and maintain this potential for 30 seconds.

(d) Repeat step C above for the following test points.

(1) Pin B of the power input connector with all other pins for the power input connector strapped to shelter facility ground.

(2) Pin C of the power input connector with all other pins of the power input connector strapped to shelter facility ground.

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(3) Pin N of the power input connector with all other pins of the power input connector strapped to shelter facility ground.

4.6.1.3 Insulation resistance. - Insulation resistance tests shall be made on each power line conductor. Test conditions, preparatory and points of measurements shall be as given in 4.6.1.2 above. The insulation resistance of all conductors with respect to ground shall not be less than 100 megohms.

4.6.1.4 Power distribution and operational functions. - Power distribution and operational function tests shall be made subsequent to the tests of 4.6.1.1 thru 4.6.1.3 with all electrical sources and loads connected. Prior to conducting the tests, the following shall be accomplished.

(1) Place all circuit breakers in the OFF position.

(2) Place all lighting controls in the OFF position.

(3) Place Blackout Bypass switch in the Bypass position.

(4) Replace all fluorescent, incandescent and neon lamps and fluorescent lamp starters.

(5) Reconnect voltmeter, frequency meter, phase indicator and ammeter circuitry and secure power distribution box cover panel.

(6) Reconnect D.C. power supplies and air conditioners

(7) Provide suitable 120 VAC test lamps or equivalent test indicators for the 120 VAC power receptacles.

(8) Apply 120/208V 3 phase, 60 hertz power to power input connector at the power entrance box. Correct phase indicator lamp shall light.

(9) Rotate voltmeter switch to position L1, L2, and L3 respectively. The voltmeter shall indicate specific voltage (approximately 120 volts) in each instance. The frequency meter shall indicate 60 hertz in each instance.

(10) Place main circuit breaker in the ON position.

(11) Verify operation of ammeter and associated circuitry for all three phases as loads are applied to system, after respective circuit breakers have been turned on.

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4.6.1.4.1 Power operational test.

a. Lighting Circuit (CB-2)

(1) Place circuit breaker CB-2 in the ON position, neon lamp above CB-2 shall light.

(2) Place fluorescent lamp switches in the ON position. All fluorescent lamps shall light.

(3) Place incandescent lamp switch in the ON position. All incandescent lamps shall light

(4) Place CB-2 in the OFF position, all fluorescent, incandescent and neon lamps mentioned in steps 1 thru 3 above shall go out.

(5) Place CB-2 in the ON position to restore light.

(6) With shelter door open, place blackout bypass switch in the Blackout position. All fluorescent and incandescent lamps shall go out.

(7) Close shelter door. All fluorescent and incandescent lamps shall light.

(8) Place fluorescent lamp switches in the OFF position. All fluorescent lamps shall go out.

(9) Place incandescent lamp switch in the OFF position.

(10) Open shelter door and restore lights by placing blackout bypass switch in the bypass position and fluorescent lamp switches in the ON position.

b. D.C. Power Supply No.1 (CB-10).

(1) Place CB-10 in the ON position, neon lamp above CB-10 shall light.

(2) Perform turn on procedures for the D.C power supply in accordance with the operator's manual.

(3) Place the D.C. power supply in the ON position and check for nominal 28 VDC between the pair of binding posts using a calibrated DC voltmeter.

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(4) Place the DC power supply in the OFF position and check for 0 volts between the pair of DC binding posts.

(5) Turn DC power supply No.1 OFF, place CB-10 in the OFF position, neon lamp above CB-10 shall go out.

c. D. C. power supply No.2 (CB-8)
Repeat steps 1 thru 5 for D. C. power supply for No. 2.

d. Heater (CB-4)

(1) Place CB-4 in the ON position; neon lamp above CB-4 shall light and test lamp in the heater receptacle shall light.

(2) Place CB-4 in the OFF position, neon lamps and test lamps mentioned above shall go out.

e. Curbside convenience outlets (CB-3).

(1) Place CB-3 in the ON position, neon lamp above CB-3 shall light, test lamps in all convenience outlets shall light.

(2) Place CB-1 in the Power Entrance Box in the ON position. Test lamp in the outside convenience outlet shall light.

(3) Place CB-1 in the OFF position. Test lamp in the outside convenience outlet shall go out.

(4) Place CB-3 in the OFF position, neon lamps and test lamps mentioned above shall go out.

f. Roadside convenience outlets (CB-9). Repeat paragraph e. above for roadside convenience outlets and CB-9.

g. Air conditioner No. 1 (CB-5)

(1) Place CB-5 in the ON position, neon lamp above CB-5 shall light.

(2) Perform turn on procedures for the air conditioner in accordance with its operator's manual. The air conditioner shall function properly.

(3) Turn air conditioner off, place CB-5 in the OFF position, neon lamp above CB-5 shall go out.

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h. Air conditioners No. 2 (CB-7).

(1) Repeat paragraph j above for air conditioner No. 2 and circuit breaker CB-7.

i. Air Compressor (CB-11).

(1) Place CB-11 in the ON position, neon lamp above CB-11 shall light.

(2) Turn on air compressor, the air compressor shall function properly.

(3) Turn air compressor off, place CB-11 in the OFF position, neon lamp above CB-11 shall go out.

j. Spare (CB-6)

(1) Place CB-6 in the on position, neon lamp above CB-6 shall light.

(2) Place CB-6 in the OFF position, neon lamp above CB-6 shall go out.

k. Main circuit breaker (CB-1)

(1) Plug all operational and test equipments (including heater) to be furnished with the facility into an appropriate electrical outlet and place all circuit breakers in the ON position.

(2) Perform turn on procedures for all equipments in accordance with their operator's manuals.

(3) With all equipments operating place main circuit breaker in the OFF position, all equipments shall cease operating and all lights, except power ON indicator lamps shall go out.

(4) Replace all circuit breakers and switches to the OFF position except fluorescent light switches. Restore lights by placing main circuit breaker to the ON position.

4.6.2 Signal Wiring. - Disconnect TA-312, for following tests:

a. Continuity

(1) continuity shall exist between each binding post outside in the power entrance box and the corresponding signal wire termination inside

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the shelter and continuity resistance shall be less than 1 ohm.

(2) Measure continuity resistance for the

Two conductors of TA-312 to phone jack (in raceway).

b. Insulation resistance

Insulation resistance test shall be made on each signal conductor. Insulation resistance of each signal conductor with respect to ground shall not be less than 100 megaohms. Measurement shall be made at each binding post.

c. Operations test:

(1) Connect two wires from a TA-312/PT telephone to binding posts 1a and 1b in the power entrance box.

(2) Connect tinned ends of telephone signal cord to the binding posts of the other TA-312/PT telephone.

(3) Operate ringer on each phone, in turn, and talk into each phone. Two way voice communication shall be possible. Repeat for spare binding posts.

4.7 High and low temperature - The assembly, when subjected to the following tests shall meet the requirement of 3.8.

a. The high temperature test according to MIL-STD-810D, Method 501.2, procedure I and II.

b. The low temperature test according to MIL-STD-810D Method 502.2, Procedure I.

4.8 Vehicular transportation test - The shelter assemblage shall be secured and blocked to prevent shifting to a Government furnished 5-ton truck ,and driven five times over sections of the Munson Test Course at Aberdeen Proving Ground, Aberdeen, MD., or approved equal, in the following order and at the specified speeds:

Coarse washboard (6-inch waves spaced 72 inches apart)	5 mph
Belgian block	20 mph
Radial washboard (2-inch to 4-inch waves)	15 mph
Two-inch washboard	10 mph

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Three-inch spaced bump 20 mph

4.9 Rail transport, impact test - The shelter assemblage secured to a flatcar shall be impacted by a loaded gondola car. The test shall be performed on a straight and level stretch of track. The test flatcar with its timber blocking and shelter tie-down slings, shall be as specified in figure 1 and shall meet the provisions of the test as specified in 4.9.1. The test flatcar (less the shelter) and the two empty flatcars coupled to it shall each weigh 45,000 to 55,000 pounds. The loaded gondola car shall meet the provisions of the test as specified in 4.9.2 and shall have a total weight of 165,000 pounds. The loaded gondola car totaling 165,000 pounds traveling at 9 miles per hour (or equivalent energy conditions acceptable to the Government Technical Representative) shall be impacted against the stationary loaded test car coupled to the empty flatcars with brakes off. The test shall consist of four impacts. The first two impacts shall be with the door end of the shelter facing the loaded gondola car; the second two impacts shall be with the door end of the shelter facing the opposite direction from the loaded gondola car. The lashing, blocking and wedges holding the shelter in place shall be tight at the start of each run. If the blocking is torn loose or the lashing is broken during impact, the run shall be repeated. After each impact the shelter assemblage shall meet the requirements of 3.5.

4.9.1 Test flatcar. - The wooden planking of the test flatcar bed shall be in good condition. All 6 x6 inch blocking shall be secured to the bed of the flat car using 10-inch spikes on 2-inch centers. The blocking along the door end and opposite end of the shelter shall fit tightly against the corner castings along the full length of the castings except as necessary to clear the towing shackles. This shall be accomplished by means of power sawed notches. The blocking along the other two sides of the shelter shall fit tightly against the corner castings, along the full length of the castings. These four blocking members shall be notched or shimmed to contact the trim along the sides of the shelter. The 4x4 inch blocking shall be nailed to the bed of the flatcar as shown in figure 1. All other blocking shall be firmly secured as shown in the figure 1. Lashing shall consist of four wire rope slings, each including 3/8 inch diameter, 6x19, flexible steel a preformed wire rope (having a breaking strength of 10,600 pounds minimum), a turnbuckle, two hooks, with thimbles and fasteners as required; the breaking strength of the sling shall be 9,000 pounds minimum. Before each impact, each sling shall be taut (600-800 pounds tension).

4.9.2 Loaded gondola car. The load in the loaded gondola car shall be one of the following:

a. A coarse size of either coal, crushed stone, gravel, concrete or metal scrap (other than sheet metal, metal shavings or other springy forms

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of metal).

b. Chunk, slab, block or plate forms of the preceding acceptable materials, constrained to prevent shifting.

4.10 Drop tests. -

4.10.1 Flat drop. - The shelter assemblage shall be lifted 18 inches from the ground and allowed to fall freely with the skids impacting onto a hard substantial concrete surface . The test shall be performed once. Upon completion of this drop, the asemblage shall meet the requiremens of 3.6.

4.10.2 Rotational Drop. - The shelter assemblage shall have one skid (or the knuckles of the skids) resting on a nominal 5-6inch high timber. The outer edge of the opposite skid (or the opposite knuckles of the skids) shall be raised 18 inches and the shelter shall be allowed to fall freely onto a hard substantial concrete surface. This impact shall be performed four times, once for each bottom edge of the shelter. On completion of these drops, the shelter assemblage shall meet the requirements of 3.6.

4.11 Air tightness test - The shelter facility when subjected to the following test, shall meet the requirements of 3.8.

a. The air tightness shall be tested by introducing clean dry air into the closed shelter through an existing appropriate opening (i.e., drain fixture) to create an internal air pressure that exceeds the external pressure by a minimum of six (6) inches of water.

b. A manometer graduated in 0.1 inches or less shall be utilized to measure this differential pressure.

c. The test apparatus shall include an air supply sufficient to maintain the required air pressure, an air flow regulator and an input air pressure gage.

d. The shelter facility shall be tested for air tightness by painting the sealed joints, seams, rivets, etc. of the external modifications with Leak-Tac Formula 372 (American gas & Chemicals, Inc., Leak-Tac Division) or equal soap and water solution.

e. To minimize air loss, temporary seals, shims and tape are permitted to seal the shelter door, blower cover and air-conditioner port opening not being tested.

4.13 Mounting insert tests. - Four of each size inserts shall be randomly

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selected for test. Each insert shall be tested with an axial load followed by a torque load in accordance with Table V. After test, the requirements of 3.9 shall be met. Failure of any insert shall be cause for rejection of the sample equipment. This test shall be conducted prior to installation of brackets, racks, etc.

TABLE V

Mounting insert loads

Insert Size	First Article and Group C Test Axial (lbs)	Torque (Inch Lbs)	Group B Test Axial (Lbs)	Torque (Inch lbs)
10-32	600	23	300	23
1/4-20	1300	60	650	60
5/16-18	2000	100	1000	100

4.14 Humidity Test - The assembly, when subjected to the humidity test of MIL-STD-810D, Method 507.2, Procedure II for Ground and Airborne Electronic Equipment, shall meet the requirement of 3.7.

4.15 Testing of solderless terminal-lug connections - Connections made with solderless terminal-lug shall be tested for conformance to 3.22.4.

4.16 Inspection for interchangeability. - The dimensions listed below shall be gauged or measured to determine compliance with the physical interchangeability requirements of 3.23. When a dimension is not within specified or design limits, it shall be considered a defect.

a. Size and mounting dimensions of brackets and mounting plates.

b. Size and mounting dimensions of racks and cabinets.

4.17 Visual and mechanical inspection - The shelter assemblage shall be examined for the defects listed in MIL-STD-252, the requirements of 3.24 and TABLE VI.

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TABLE VI

Classification of Visual and Mechanical Defects

Classification		Defects
Major	1.	Mounting plates missing or mislocated.
	2.	Aircraft loading data plate, instruction or nameplate omitted, incorrect or illegible
	3.	Welds-cracked or porous. Inspection of welded assemblies, such as equipment racks entrance boxes, etc, should be made prior to final paint application.
	4.	Sealer improperly applied (holes, separations, or lack of adhesion).
	5.	Mounting inserts missing or mislocated
	6.	Burrs or sharps edges in wire duct not removed.
	7.	Equipments or bracketry loose or missing.
	8.	Doors, etc.
	9.	Electrical receptacles, switches, jacks, connectors, wire ducts, or other electrical components loose, improperly located or inoperative.
	10.	Locking or holding devices missing or inoperative.
	11.	Gaps between sections of wire duct exceed 1/16.
	12.	The blower housing or air conditioner mount does not align properly with the mounting inserts of the shelter facility.
	13.	Ground stud not properly sealed.
	14.	Bulkhead power and signal connector fasteners not sufficiently tight.
	15.	Electrical schematics missing.

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Minor	1.	Doors not easily operable.
	2.	Fastening devices difficult to operate.
	3.	Finish-abrasions or scratches.
	4.	Burrs and sharp edges on bracket removed.
	5.	Screws loose.
	6.	Gaskets, not completely and securely adhered.
	7.	Electrical schematics not properly adhered to wall.
	8.	Cable assemblies not dressed properly.

4.18 Packaging inspection. - Packaging inspection requirements specified herein are classified as follows:

a. First Article Inspection of Packaging.

b. Quality Conformance inspection of Packaging.

4.18.1 First Article Inspection Packaging. Unless otherwise specified in the contract, First Article Inspection of Pakaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.18.2 Quality Conformance Inspection of Packaging.

4.18.2.1 Materials inspection. - All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.18.2.2 Preservation inspection. - Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.18.2.3 Packing inspection. - Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in Table VII, "PACKING INSPECTION PROVIISIONS". Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and Acceptable Quality level of 4.0 percent defective.

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TABLE VII

PACKING INSPECTION PROVISIONS.

No.	CHARACTERISTIC	METHOD OF INSPECTION
101	Intermediate container not as specifed	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

4.19 Deleted

4.20 Acoustical noise test. - The shelter assemblage shall be located in an area remote from excessive machinery, vehicle and other noise operating sources to reduce ambient noise to a minimum, with the shelter door closed. Only necessary test personnel shall be present in the shelter. Using Octave Band Analyzer, General Radio Corp., Model 1558A, or equal, and Microphone, Peizoelectric, General Radio Corp. Model 1560P6 or equal , record ambient noise levels over the frequeny ranges specified in 3.11. Operate all equipment components and repeat noise level recording at the same frequency levels as above. The noise levels, corrected from ambient, shall meet the requirements of 3.11. Test measurements of MIL-STD-1474B (MI) for steady-state noise in personnel occupied

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areas, Para 5.1.

4.21 Cable Assemblages, power and audio frequency. - Each cable assembly shall be tested for continuity, dielectric, insulation resistance and pull-out per MIL-C-3885 (See 3.3.3.)

4.22 Tropicalization of material. - All materials of construction shall be in accordance with the equipment drawings and shall be Fungus-inert. Substitution of any materials other than specified must be approved by the contracting officer. The material will only be approved after TOAD tests the material for Fungus-Resistance satisfactorily in accordance with requirement 508.3 of MIL-STD-810D.

4.23 System Safety Inspection. - An equipment visual inspection shall be performed to show that all the requirements of Para 3.16 have been met (use Form SEL-1183). Inability to pass this inspection will constitute test failure.

5. PACKAGING

5.1 Packaging requirements. - The requirements for packaging shall be in accordance with MIL-S-55507.

6. NOTES:

6.1 Intended use. - Electronic shops are used to provide backup maintenance support for electronic equipment components. Specific use is as follows:

Electronic Shop, Shelter Mounted AN/MSM-108( ) is used as the repair or maintenance shop for accomplishing the maintenance functions.

6.2 Ordering data - Procurement documents should specify the following:

a. Title, number,and date of this specification and any amendment thereto.

b. First article and Group C inspection of the Shelter assemblage requires use of contractor fabricated dummy loads or Government furnished live loads (operational equipment) as specified in the bid request and contract.

c. A list of Government furnished equipments required as part of the AN/MSM-108.

d. A list of equipments to be Government loaned for testing purposes.

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e. Level A or B preservation and packing (see section 5)

f. When first article rough handling tests are not required.

g. When first article packaging inspection reports require acquisition activity approval prior to production unit packing.

6.3 Level B preservation. - When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

6.4 Environmental. - Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity; for recommended disposability methods.

6.5 Subject Term (key word) listing.

Article
Assemblage
Classification of inspections
Electrical installation
Inspections
Operations test
Packing inspection provisions
Power distribution
Power operational test
Quality assurance provisions
Quality conformance
Reinspection
Requirements
Specifications
Standards
Statement of Work
Test
Verification

Custodian:

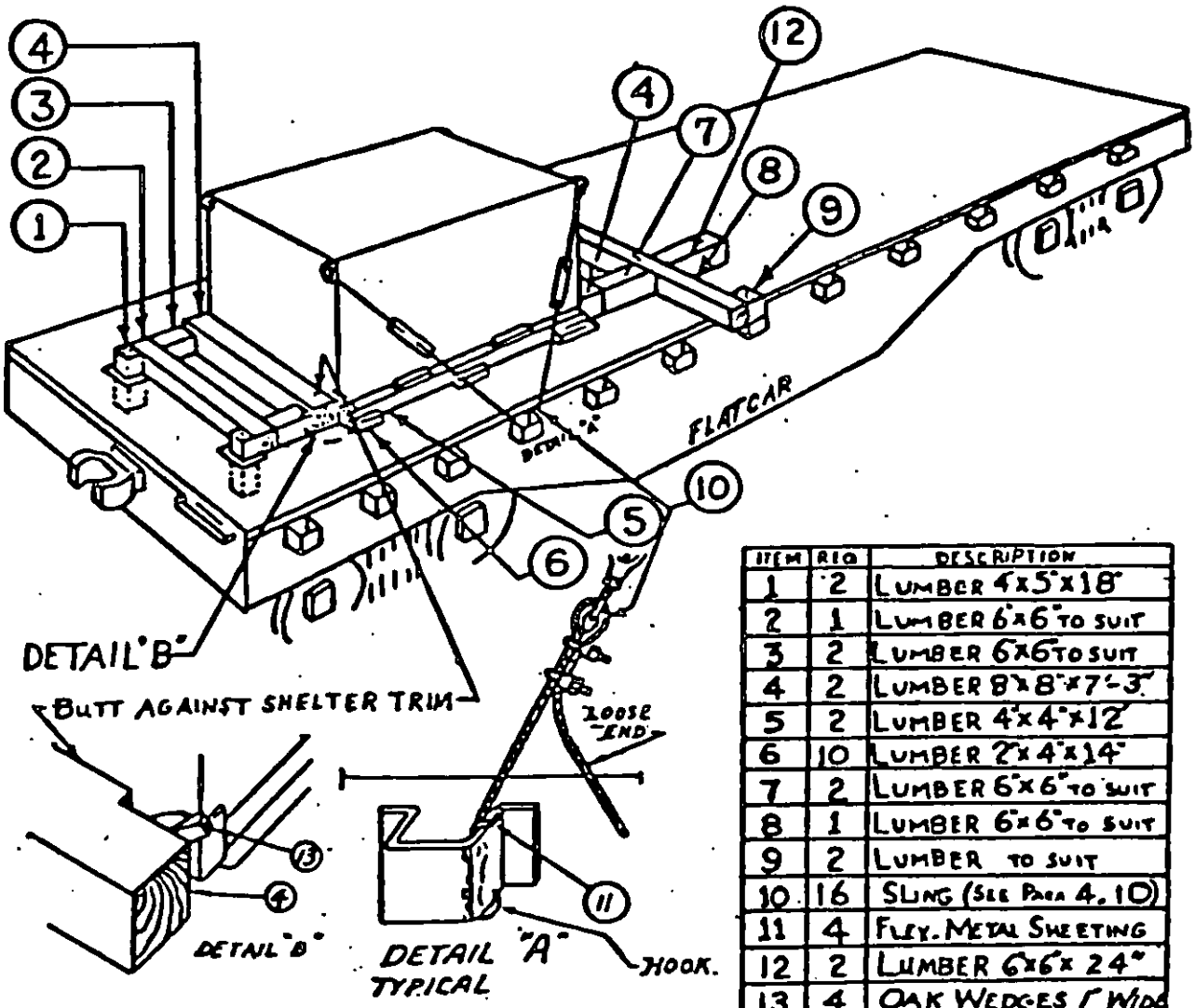
ARMY-CR

Preparing Activity

ARMY-CR

Project 5410-A311

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ITEM	REQ	DESCRIPTION
1	2	LUMBER 4"x5"x18"
2	1	LUMBER 6"x6" TO SUIT
3	2	LUMBER 6"x6" TO SUIT
4	2	LUMBER 8"x8"x7'-3"
5	2	LUMBER 4"x4"x12"
6	10	LUMBER 2"x4"x14"
7	2	LUMBER 6"x6" TO SUIT
8	1	LUMBER 6"x6" TO SUIT
9	2	LUMBER TO SUIT
10	16	SLING (SEE PARA 4.10)
11	4	FLEX. METAL SHEETING
12	2	LUMBER 6"x6"x 24"
13	4	OAK WEDGES 1' WIDE

NOTES:
a. The blocking lumber shall be notched to accomodate towing eyes, and bolts. Wedge item 13 between towing corner forging and item 4 notch (Detail B)
b. The blocking lumber shall be rigidly fastened to the floor of the flatcar.

FIGURE 1

SHELTER SECURED ON FLATCAR

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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